IE 7615- Deep Learning for AI

DISCRIMINATIVE PROJECT MILESTONE 1 REPORT

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1. Dataset and Problem Overview

For this milestone we worked with a 73-class image classification problem. The whole class, across both sections, photographed one unique household or personal object each, around 100 images per object, including a small number of background-only reference shots, and we merged all the individual datasets into one shared class dataset.

After merging, we removed the background-only images and a small number of byte-identical duplicates, leaving 6,948 usable images across 73 object classes (object IDs OBJ001 through OBJ073). We split the merged dataset into training, validation, and test sets using a stratified 70/15/15 split with a fixed random seed of 42, so every class is proportionally represented in all three splits. The resulting split contains 4,863 training images, 1,042 validation images, and 1,043 test images.

The task is single-label multi-class classification: given a photo, predict which of the 73 registered objects it shows. The dataset is intentionally hard for this milestone, since there's a relatively small number of training images per class, roughly 65 to 70 on average, paired with a large number of classes. That's what motivates comparing a model trained from scratch against several pretrained, transfer-learning approaches.

2. Models Tested

We evaluated five convolutional architectures, several in both a frozen (feature-extraction) and fully fine-tuned configuration, for nine total experimental runs:

1. Simple CNN (from scratch): three convolutional blocks (32, 64, 128 channels) with batch normalization, max-pooling, and a dense classification head with dropout. Trained with no pretrained weights, serving as the baseline.
2. ResNet50: frozen and fine-tuned, pretrained on ImageNet.
3. EfficientNet-B0: frozen and fine-tuned, pretrained on ImageNet.
4. MobileNetV2: frozen and fine-tuned, pretrained on ImageNet.
5. DenseNet121: fine-tuned only, pretrained on ImageNet.
6. ConvNeXt-Tiny: fine-tuned only, pretrained on ImageNet.

Frozen runs keep all pretrained backbone weights fixed and train only the replaced classification head. Fine-tuned runs update the entire network. Comparing the two regimes for the same architecture isolates the effect of transfer learning versus simple feature re-use on this dataset.

Shared Experimental Setup

- Image size: 224 × 224, RGB
- Batch size: 32
- Optimizer: Adam (learning rate 1e-3 for scratch/frozen runs, 1e-4 for fine-tuned runs)
- Up to 20 epochs per run, with early stopping (patience = 5 epochs on validation loss)
- Augmentation (training only): random resized crop, horizontal flip, rotation (±15°), color jitter (brightness/contrast/saturation/hue)
- Same train/validation/test split and random seed used across all nine runs for a fair comparison

3. Experimental Results

Full results for all nine runs, sorted by test accuracy:

Model	Acc.	Prec.	Recall	F1	Params	Train (s)	Infer. (ms/img)
ResNet50 (fine-tuned)	0.9847	0.9862	0.9846	0.9850	23.66M	1169.2	9.41
ConvNeXt-Tiny (fine-tuned)	0.9818	0.9825	0.9816	0.9816	27.88M	1931.2	5.90
EfficientNet-B0 (fine-tuned)	0.9799	0.9813	0.9799	0.9799	4.10M	1053.0	12.52
ResNet50 (frozen)	0.9655	0.9687	0.9657	0.9654	23.66M	1033.7	6.32
MobileNetV2 (fine-tuned)	0.9636	0.9673	0.9639	0.9633	2.32M	1010.5	6.10
DenseNet121 (fine-tuned)	0.9607	0.9645	0.9609	0.9607	7.03M	1045.9	16.32
EfficientNet-B0 (frozen)	0.9473	0.9493	0.9478	0.9470	4.10M	971.7	8.30
MobileNetV2 (frozen)	0.9386	0.9417	0.9392	0.9384	2.32M	955.6	5.61
Simple CNN (scratch)	0.4190	0.4273	0.4196	0.3936	0.64M	962.0	1.23

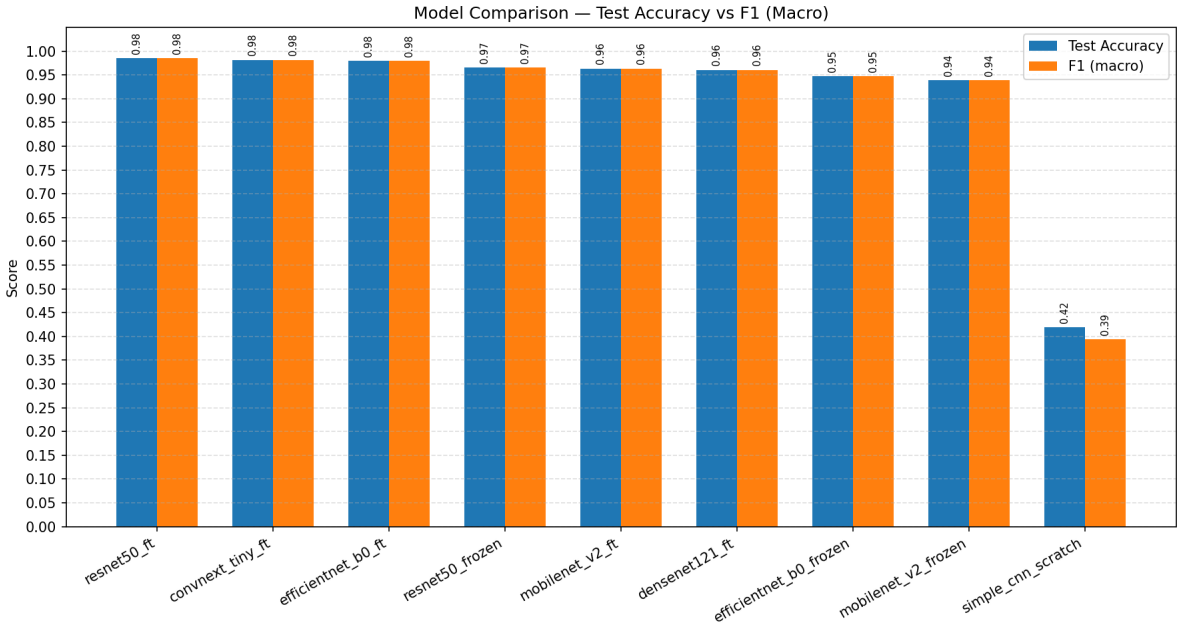


Figure 1. Test accuracy and macro F1 across all nine model configurations.

4. Comparison and Analysis

Transfer learning vs. training from scratch

The Simple CNN trained from scratch reached only 41.9% test accuracy, dramatically below every pretrained model. With roughly 65 to 70 training images per class, there just isn't enough data for a randomly-initialized network to learn useful visual features on its own. This result is direct evidence that pretrained backbones are necessary at this dataset size, not merely helpful.

Frozen backbone vs. fine-tuned

Every architecture tested in both regimes showed a clear, consistent gain from fine-tuning over freezing the backbone:

- ResNet50: 96.55% frozen, 98.47% fine-tuned, a gain of 1.92 points
- EfficientNet-B0: 94.73% frozen, 97.99% fine-tuned, a gain of 3.26 points
- MobileNetV2: 93.86% frozen, 96.36% fine-tuned, a gain of 2.50 points

This shows that while ImageNet-pretrained features are a strong starting point, letting the backbone adapt to this dataset's specific objects and photo conditions, things like varied lighting, backgrounds, and different phone cameras, consistently improves performance regardless of architecture.

Accuracy vs. efficiency trade-off

ResNet50 (fine-tuned) achieved the highest raw accuracy, 98.47%, and F1 of 0.9850, but at 23.66M parameters it's the second-largest model tested. EfficientNet-B0 (fine-tuned) trails it by only 0.48 accuracy points, 97.99%, while using roughly one-sixth of the parameters, 4.10M, which makes it a strong candidate where model size or deployment cost matters. ConvNeXt-Tiny (fine-tuned) is competitive in accuracy, 98.18%, but had the longest training time of any model, 1931.2 seconds, for only a marginal accuracy gain over EfficientNet-B0.

MobileNetV2 and DenseNet121 both performed respectably too, 96.36% and 96.07% fine-tuned accuracy, with substantially fewer parameters than ResNet50. That confirms lightweight architectures remain viable on this task once fine-tuned, even if they don't top the leaderboard.

Inference speed

Inference time per image ranged from 1.23 ms for the Simple CNN, reflecting its much smaller size and lower accuracy, up to 16.32 ms for DenseNet121. Inference speed didn't track parameter count or accuracy in any simple way: ConvNeXt-Tiny, despite having the most parameters of any model, had one of the fastest inference times at 5.90 ms, while DenseNet121 was the slowest despite a moderate parameter count, likely because its densely-connected layer structure adds more memory-access overhead per forward pass.

Training curves

Loss and accuracy curves (train vs. validation) for the overall best model and the two architectures focused on in more detail below, alongside the from-scratch baseline for contrast:

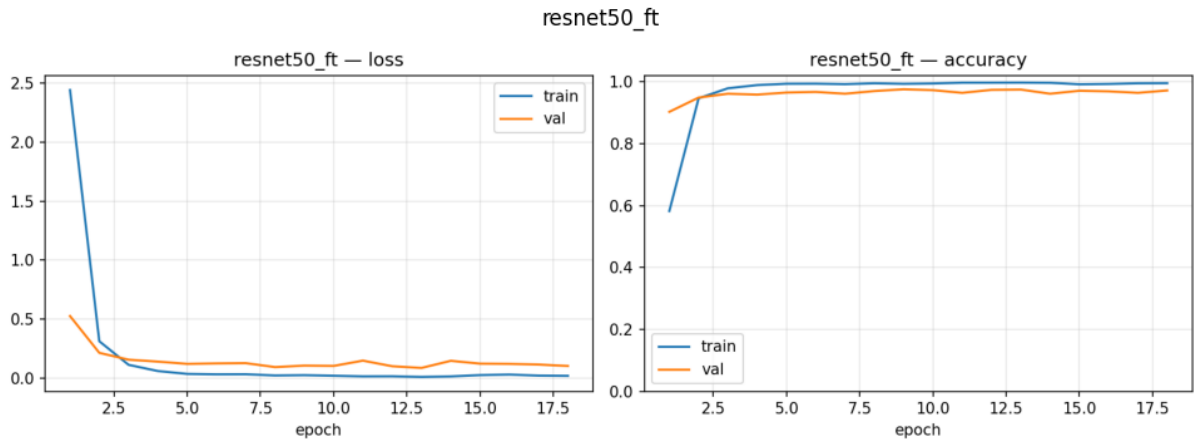


Figure 2. ResNet50 (fine-tuned), loss and accuracy curves. Best overall model.

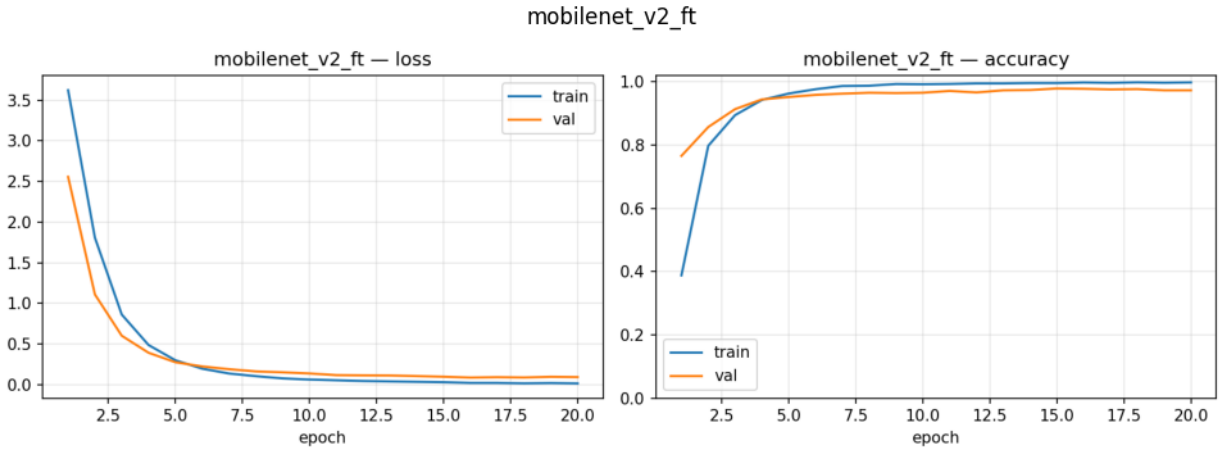


Figure 3. MobileNetV2 (fine-tuned), loss and accuracy curves.

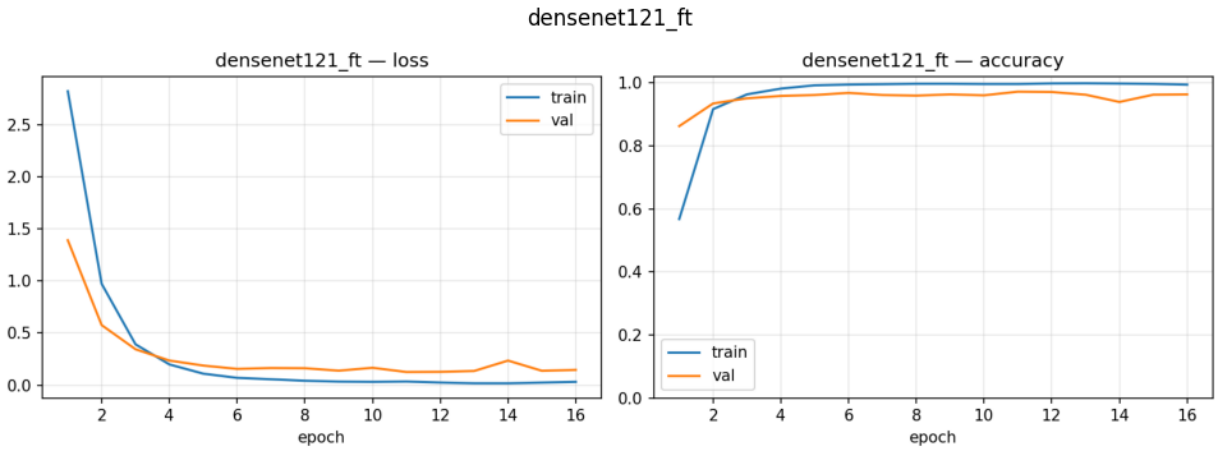


Figure 4. DenseNet121 (fine-tuned), loss and accuracy curves.

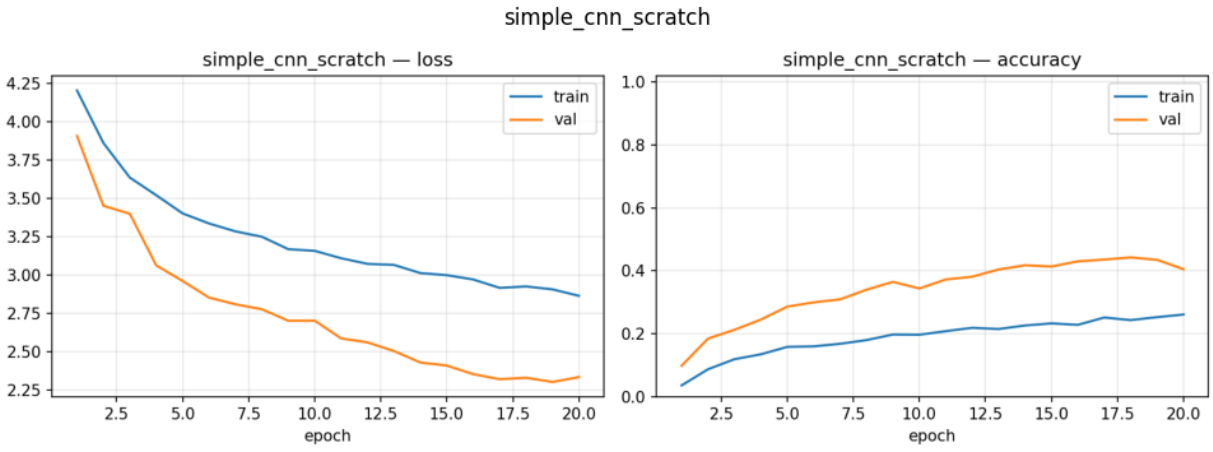


Figure 5. Simple CNN (from scratch), loss and accuracy curves. Validation accuracy plateaus far below the pretrained models, visually confirming the scratch vs. transfer-learning gap.

Error analysis

We took a closer look at the confusion matrices for the fine-tuned MobileNetV2 and DenseNet121 models, and found that misclassifications were sparse and mostly isolated to single test images rather than any systematic class confusion. The clearest repeated pattern involved a power bank (OBJ057) being confused with both a notebook (OBJ003) and a Marshall speaker (OBJ009) across both models, which makes sense given they share a rectangular, dark, matte-finish form factor. A sloth plushie (OBJ071) and a sleep mask (OBJ060) were also confused by both models, likely due to similar soft fabric texture and compact shape at certain angles or distances. Most other errors looked like they came from individual difficult photos, an unusual angle, lighting, or distance, rather than genuine visual similarity between object classes.

5. Final Model Selection and Justification

We're picking ResNet50 (fine-tuned) as the best-performing model for this dataset. It has the highest test accuracy, 98.47%, and F1-score, 0.9850, of all nine configurations we tested, with a moderate inference cost of 9.41 ms/image that's well within the range of the other fine-tuned models.

This choice holds up well: ResNet50 clearly separates from every other model on the leaderboard, including the next-closest fine-tuned alternatives, ConvNeXt-Tiny and EfficientNet-B0, and its training cost is comparable to or lower than most other fine-tuned options, 1169.2 seconds, cheaper than both ConvNeXt-Tiny and the frozen ResNet50 run. Where parameter count or inference latency is a hard constraint, for example on-device or embedded deployment, we'd recommend EfficientNet-B0 (fine-tuned) as a strong secondary choice: it's within 0.5 accuracy points of ResNet50 while using roughly one-sixth the parameters.

6. Limitations and Next Steps

- Small per-class sample size: roughly 65 to 70 training images per class across 73 classes leaves limited room to assess true generalization, particularly for the from-scratch baseline.
- Photo inconsistency: images were captured independently across the class using different phones, lighting, and backgrounds, which introduces variance that a production dataset would typically control for.
- Look-alike objects: a small number of object pairs, like the power bank vs. notebook/speaker, or the plushie vs. sleep mask, share enough visual similarity to cause occasional misclassification even in the best-performing models.
- Background-only images were excluded from this milestone's classification task. They're reserved for Milestone 2, which extends this work to multi-object images and YOLOv8-based object detection using transfer learning from the classification backbones evaluated here.

7. Code Repository

All code for every model tested (the custom CNN and all transfer-learning runs), with a README explaining the repository and how to run it, is publicly available on GitHub:

https://github.com/sankalphegde/ie7615-discriminative-project